

Phase 1

The Warehouse Digital Twin Platform is an AI-powered operational intelligence system that transforms existing CCTV infrastructure into a real-time virtual representation of warehouse operations.

Using Computer Vision, Tracking, Event Intelligence, and Digital Twin technologies, the platform continuously monitors warehouse activities, analyzes operational behavior, detects safety risks, measures productivity, and generates actionable insights for managers.

Unlike traditional CCTV systems that only record video for later review, the Digital Twin continuously converts video into operational intelligence that can be used for monitoring, optimization, prediction, and simulation.

The platform is designed to help warehouse operators:

- Improve operational visibility
 - Increase warehouse throughput
 - Reduce safety incidents
 - Optimize loading dock performance
 - Improve workforce productivity
 - Enable predictive decision-making
-

Vision Statement

Current State

Most warehouses have dozens of CCTV cameras generating thousands of hours of video every day.

However:

- Managers cannot monitor all cameras simultaneously.
- Operational bottlenecks remain hidden.
- Safety violations are discovered too late.
- Labor inefficiencies are difficult to measure.
- Decisions are reactive rather than proactive.

Future State

The Digital Twin creates a continuously updated virtual model of warehouse operations.

Managers can instantly see:

- Where workers are located
- Where forklifts are operating
- Which zones are congested
- Which docks are occupied
- Where safety risks exist
- How efficiently the warehouse is operating

The result is a shift from video surveillance to operational intelligence.

Business Objectives

Objective 1: Real-Time Warehouse Visibility (MVP)

Business Problem

Warehouse managers lack real-time visibility into operations.

They cannot effectively answer:

- How many workers are currently active?
- Which areas are congested?
- Where are forklifts operating?
- Which zones are underutilized?

Digital Twin Solution

The platform creates a live operational map of the warehouse.

Features:

- Live warehouse layout
- Zone occupancy tracking
- Worker distribution monitoring
- Forklift tracking
- Congestion detection
- Zone utilization analysis

KPIs

- Occupancy Accuracy >95%
- Alert Response Time
- Time-to-Detect Operational Issues
- Average Zone Utilization

Business Impact

Provides immediate operational awareness and enables faster decision-making.

Objective 2: Safety Intelligence

Business Problem

Warehouses face ongoing safety risks:

- PPE violations
- Unauthorized access
- Pedestrian-forklift interactions
- Unsafe movement patterns

Traditional CCTV systems rely on manual monitoring.

Digital Twin Solution

AI continuously monitors safety conditions.

Features:

- Helmet Detection
- Safety Vest Detection
- Restricted Zone Monitoring
- Forklift-Pedestrian Proximity Detection
- Near-Miss Analysis
- Safety Event Logging

KPIs

- PPE Compliance %
- Safety Violations per Day
- Near Miss Count

- Restricted Area Violations

Business Impact

Reduces workplace incidents and improves compliance.

Objective 3: Loading Dock Optimization

Business Problem

Loading docks are often bottlenecks.

Common issues:

- Truck queues
- Dock congestion
- Long turnaround times
- Detention penalties

Digital Twin Solution

The platform continuously analyzes dock operations.

Features:

- Dock Occupancy Monitoring
- Truck Queue Tracking
- Dock Utilization Analysis
- Loading Duration Tracking
- Turnaround Time Monitoring

KPIs

- Dock Utilization %
- Truck Waiting Time
- Truck Turnaround Time
- Dock Dwell Time

Business Impact

Improves throughput and reduces logistics costs.

Objective 4: Workforce Productivity Optimization

Business Problem

Labor costs are among the largest warehouse expenses.

Managers often lack objective visibility into workforce efficiency.

Digital Twin Solution

The system generates workforce intelligence.

Features:

- Worker Density Heatmaps
- Activity Monitoring
- Zone Productivity Analysis
- Idle Time Detection
- Workforce Distribution Analysis

KPIs

- Worker Utilization
- Labor Productivity
- Picks per Hour
- Active vs Idle Time

Business Impact

Improves labor allocation and operational efficiency.

Objective 5: Predictive Warehouse Intelligence

Business Problem

Most warehouse decisions are reactive.

Issues are addressed only after they occur.

Digital Twin Solution

The Digital Twin evolves from monitoring to prediction.

Features:

- Occupancy Forecasting
- Congestion Prediction
- Throughput Forecasting
- Resource Planning Recommendations
- Risk Prediction

KPIs

- Prediction Accuracy
- Throughput Increase
- Resource Utilization
- Congestion Reduction

Business Impact

Enables proactive operations management.

Solution Architecture

The solution follows an Edge AI + Event Intelligence architecture.

Core Design Principle

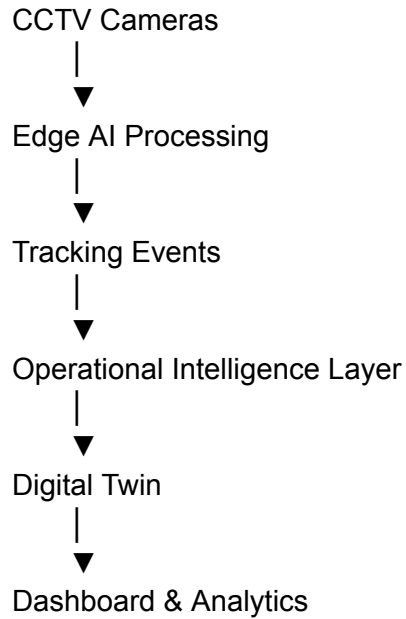
Video remains inside the warehouse.

Only structured events are transmitted.

This provides:

- Lower bandwidth costs
 - Better privacy
 - Faster processing
 - Easier scalability
-

System Architecture

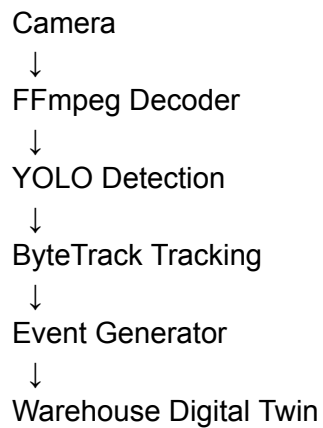


Phase 1 Architecture (Single Camera)

Objective

Validate core warehouse analytics pipeline.

Flow



Component Responsibilities

Camera

Captures warehouse activity.

Examples:

- Workers
 - Forklifts
 - Trucks
 - Loading docks
-

FFmpeg

Converts RTSP video streams into frames.

Responsibilities:

- Stream decoding
 - Frame sampling
 - Resolution normalization
-

YOLO Detection

Detects:

- Person
- Forklift
- Truck
- Pallet
- Safety Equipment

Output:

```
{  
  "object": "forklift",  
  "bbox": [x,y,w,h]  
}
```

ByteTrack

Maintains object identity.

Example:

Forklift 12

remains

Forklift 12

across multiple frames.

Enables:

- Dwell time
- Occupancy
- Path analysis

Event Generator

Converts detections into warehouse events.

Examples:

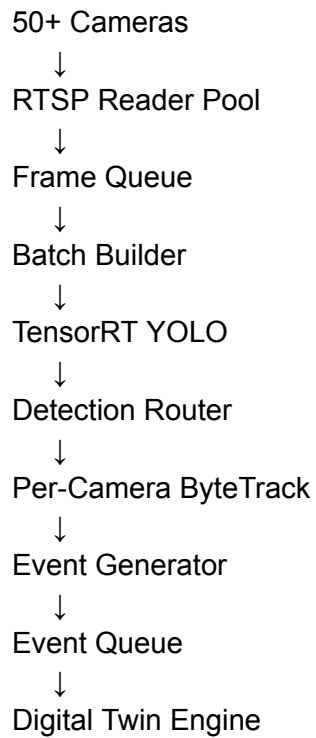
```
{  
  "event": "zone_enter"  
}  
{  
  "event": "forklift_near_worker"  
}  
{  
  "event": "dock_occupied"  
}
```

Phase 2 Architecture (Multi-Camera Warehouse)

Objective

Create a complete warehouse-wide Digital Twin.

Multi-Camera Processing Pipeline



Digital Twin Engine

This becomes the heart of the platform.

The Digital Twin maintains:

Warehouse State

Current occupancy.

Workforce State

Current worker locations.

Equipment State

Current forklift locations.

Dock State

Current loading dock utilization.

Safety State

Current safety violations.

Event Intelligence Layer

The Event Intelligence Layer transforms raw tracking data into operational events.

Examples:

Occupancy Events

```
{  
  "zone": "Picking Area",  
  "occupancy": 23  
}
```

Congestion Events

```
{  
  "zone": "Packing Area",  
  "severity": "High"  
}
```

Safety Events

```
{  
  "event": "No Helmet"  
}
```

Dock Events

```
{  
  "dock": "Dock-4",  
  "status": "Occupied"  
}
```

Dashboard Modules

Live Warehouse Twin

Interactive warehouse map.

Displays:

- Workers
 - Forklifts
 - Trucks
 - Occupancy
-

Safety Dashboard

Displays:

- PPE Compliance
 - Near Misses
 - Restricted Zone Access
-

Loading Dock Dashboard

Displays:

- Dock Occupancy
 - Truck Queue Length
 - Turnaround Time
-

Workforce Dashboard

Displays:

- Worker Density
 - Activity Heatmaps
 - Utilization Metrics
-

Analytics Dashboard

Displays:

- Throughput
 - Occupancy Trends
 - Historical Performance
-

Phase 3 Architecture (Predictive Digital Twin)

Objective

Move from visibility to prediction.

New Components

Historical Event Store



Feature Engineering



Prediction Models



Digital Twin Forecast Engine

Predictive Models

Occupancy Prediction

Forecast future occupancy.

Congestion Prediction

Predict bottlenecks before they occur.

Throughput Forecasting

Estimate future warehouse throughput.

Resource Planning

Recommend staffing levels.

Future Vision: Autonomous Warehouse Intelligence

The final evolution of the Digital Twin is a self-learning warehouse intelligence platform.

Monitor



Analyze



Predict



Recommend



Simulate

Eventually managers can ask:

"What happens if I move 3 workers from Zone A to Zone B?"

or

"Which loading dock configuration will reduce truck waiting time by 20%?"

and the Digital Twin can simulate the outcome before changes are made.

Poc diagram

EDGE TIER — one GPU box per facility (near cameras)

CCTV cameras
(RTSP)



FFmpeg decode
tiered 1–5 fps



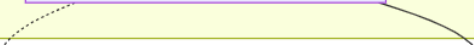
YOLO detect
(TensorRT)

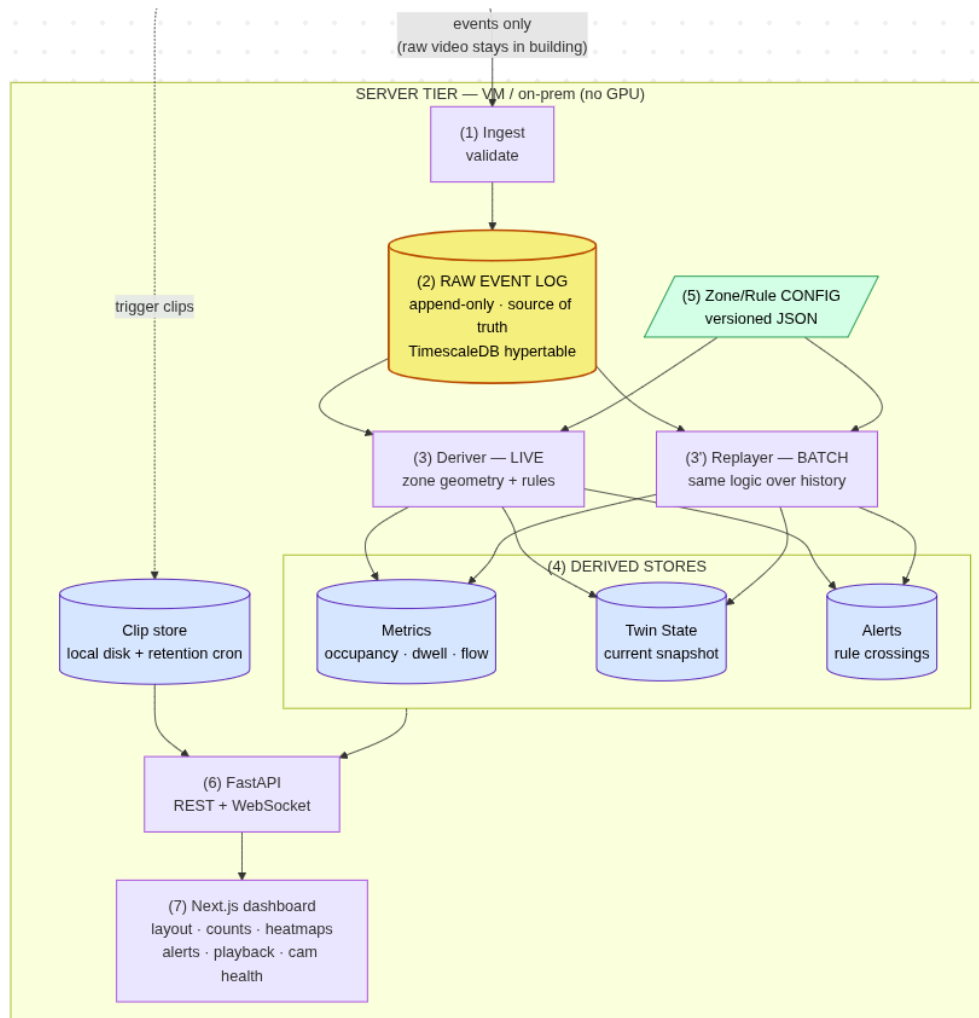


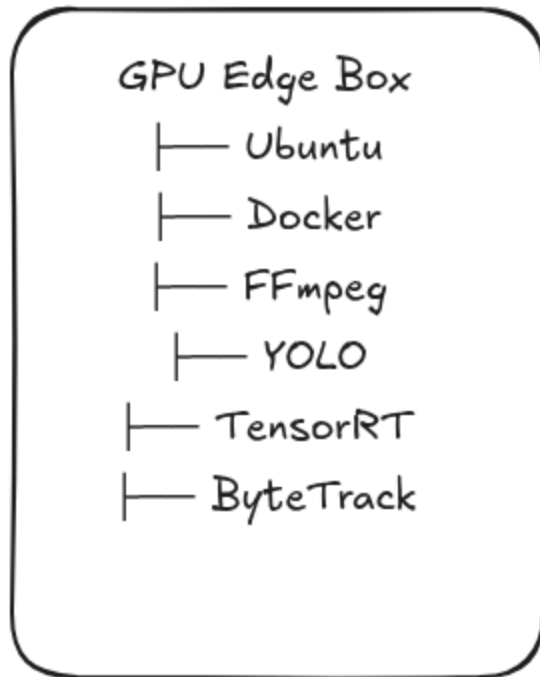
ByteTrack
single-camera tracking



Edge emitter
produces TRACK EVENTS







POC Roadmap

POC V1

Deliver:

- ✓ Warehouse Layout
- ✓ Camera Integration
- ✓ Person Detection
- ✓ Forklift Detection
- ✓ Zone Occupancy
- ✓ Heatmaps
- ✓ Congestion Detection
- ✓ Live Dashboard

POC V2

Deliver:

- ✓ PPE Monitoring
 - ✓ Restricted Zone Detection
 - ✓ Dock Monitoring
 - ✓ Historical Analytics
 - ✓ Event Playback
-

POC V3

Deliver:

- ✓ Occupancy Forecasting
 - ✓ Congestion Prediction
 - ✓ Throughput Forecasting
 - ✓ Simulation Engine
 - ✓ Resource Planning
-

Management Summary

The Warehouse Digital Twin transforms existing CCTV infrastructure into a real-time operational intelligence platform. By combining AI-powered computer vision, tracking, event intelligence, and digital twin technologies, the solution provides live visibility, safety monitoring, workforce analytics, loading dock optimization, and predictive operational intelligence. The platform enables warehouse operators to improve throughput, reduce operational costs, enhance safety, and make data-driven decisions using a continuously updated virtual representation of warehouse operations.

Phase 2 and 3 will be involving robots and autonomous warehouses